

An anime-style illustration of a young woman with long black hair and a small earring, looking down with a sad expression. She is wearing a dark jacket with red accents. The background is a rainy city street at night, with blurred lights and buildings. The entire scene is framed within a circular vignette.

Relational Database for Anime

Team Members: Devanshi Parmar, Lili Gibson, Josh McDaniel

ITSC 3160 Database Design and Implementation

Instructor: Azhar Ahamad

April 29, 2026



Project Overview

Our project is a relational database designed to store and manage anime information, which allows efficient querying of relationships between anime, their characters, and the people involved in the production.

Objectives:

- Organize anime - related data into structured tables with clear relationships
- Enable efficient retrieval of information

Scope:

- Anime Catalog: Titles, release year, season, episode count
- Characters & Voice Actors: Character details and their voice actor mappings
- Production Details: Studios, genres, demographics, and source material
- Statistics: Scores, rankings, popularity, and favorites





Requirements Gathering

Data Requirements

- Anime (title, release year, season, episodes)
- Characters (name, role, age, favorites)
- Voice Actors (name, birthday, language)
- Studios (name, creation date)
- Genres (genre name)
- Production (studio, genre, demographic, source)
- Anime Stats (score, ranking, popularity)

Functional Requirements

- Add/update/delete anime, characters, voice actors, studios
- Link characters to anime and voice actors
- Assign genres and studios to each anime
- Retrieve data (ex. anime by genre, characters by anime)
- Track and display anime statistics (score, ranking, popularity)

Security/Integrity Requirements

- Use primary keys for unique identification
- Use foreign keys to maintain relationships between tables
- Prevent duplicate or inconsistent data entries
- Restrict invalid inputs (ex. missing required fields)



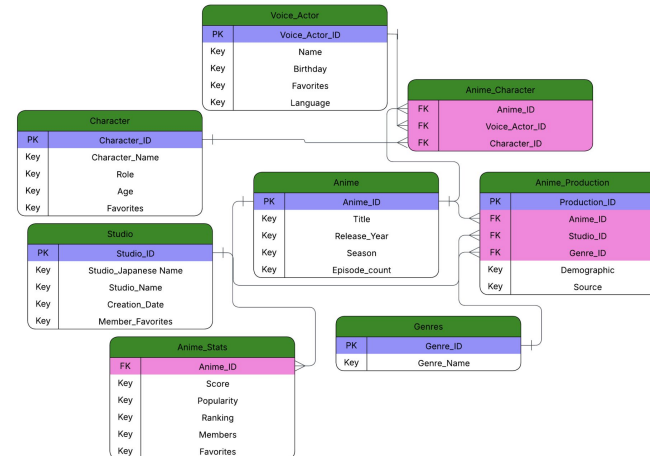
Conceptual Design (ER Diagram)

Core Entities & Relationships

- **Anime:** Central entity storing title, release year, season, and episodes
- **Character:** Represents anime characters with roles and attributes
- **Voice_Actor:** Stores voice actor details linked to characters
- **Studio:** Contains production studio information
- **Genres:** Defines anime categories
- **Anime_Production:** Connects anime with studios, genres, and production details
- **Anime_Character:** Bridge table linking anime, characters, and voice actors
- **Anime_Stats:** Stores ratings, popularity, and rankings

Key Relationships

- One **Anime** → many **Characters** (via Anime_Character)
- One **Character** → many **Voice Actors** (and vice versa)
- One **Anime** → one/many **Studios & Genres** (via Anime_Production)
- One **Anime** → one set of **Stats**



[Link to Lucidchart](#)



Logical Design (Relational Model)

	Anime
Primary Key	Anime_ID
Key	Title
Key	Release_Year
Key	Season
Key	Episode_Count

	Character
Primary Key	Character_ID
Key	Character_Name
Key	Role
Key	Age
Key	Favorites

	Studio
Primary Key	Studio_ID
Key	Studio_Name
Key	Studio_Japanese_Name
Key	Creation_Date
Key	Member_Favorite

	Anime_Production
Primary Key	Production_ID
Foreign Key	Anime_ID
Foreign Key	Studio_ID
Foreign Key	Genre_ID
Key	Demographic
Key	Source

	Voice_Actor
Primary Key	Voice_Actor_ID
Key	Name
Key	Birthday
Key	Favorites
Key	Language

	Anime_Character
Foreign Key	Anime_ID
Foreign Key	Character_ID
Foreign Key	Voice_Actor_ID

	Genres
Primary Key	Genre_ID
Key	Genre_Name

- 1NF (First Normal Form):
 - All tables have atomic values (no repeating groups or multi-valued attributes)
- 2NF (Second Normal Form):
 - All non-key attributes fully depend on the primary key
 - Bridge table (**Anime_Character**) resolves many-to-many relationships
- 3NF (Third Normal Form):
 - No transitive dependencies (non-key attributes depend only on PK)
 - EX: Studio details stored only in **Studio**, not repeated in Anime
 - Genre and production details separated into **Anime_Production**

Physical Model 1

[Link to Physical Model Google Docs](#)

Sample Table and Data

Anime Table

	ANIME_ID	TITLE	RELEASE_YEAR	SEASON	EPISODE_COUNT
1	1	Sousou no Frieren	2023	Fall	28
2	2	Steel Ball Run: JoJo r	(null)	TBA	0
3	3	Fullmetal Alchemist:	2009	Spring	64
4	4	Chainsaw Man Movie	2025	Movie	1
5	5	Steins;Gate	2011	Spring	24
6	6	Shingeki no Kyojin Se	2019	Spring	10
7	7	Gintama*	2015	Spring	51
8	8	Gintama: The Final	2021	Movie	1
9	9	Hunter x Hunter (201	2011	Fall	148
10	10	One Piece Fan Letter	2024	Special	1

```
CREATE TABLE anime (  
    anime_id NUMBER PRIMARY  
    KEY,  
    title VARCHAR2(255),  
    release_year NUMBER,  
    season VARCHAR2(20),  
    episode_count NUMBER  
);
```

SQL Queries

```
INSERT INTO ANIME VALUES (1,Sousou no  
Frieren',2023,'Fall',28);
```

```
INSERT INTO ANIME VALUES (2,Steel Ball  
Run: JoJo no Kimyou na Bouken',NULL,TBA',0);
```

```
INSERT INTO ANIME VALUES (3,Fullmetal  
Alchemist: Brotherhood',2009,'Spring',64);
```

```
INSERT INTO ANIME VALUES (4,'Chainsaw  
Man Movie: Reze-hen',2025,'Movie',1);
```

```
INSERT INTO ANIME VALUES  
(5,'Steins;Gate',2011,'Spring',24);
```

```
INSERT INTO ANIME VALUES (6,'Shingeki no  
Kyojin Season 3 Part 2',2019,'Spring',10);
```

```
INSERT INTO ANIME VALUES (7,'Gintama*  
'',2015,'Spring',51);
```

```
INSERT INTO ANIME VALUES (8,'Gintama:  
The Final',2021,'Movie',1);
```

```
INSERT INTO ANIME VALUES (9,'Hunter x  
Hunter (2011)',2011,'Fall',148);
```

```
INSERT INTO ANIME VALUES (10,'One Piece  
Fan Letter',2024,'Special',1);
```

SQL Queries

```
INSERT INTO CHARACTERS (CHARACTER_ID, NAME,  
AGE, FAVORITES, ROLE)
```

```
VALUES (1, 'Fern', 16, 5663, 'Main');
```

```
INSERT INTO CHARACTERS (CHARACTER_ID, NAME,  
AGE, FAVORITES, ROLE)
```

```
VALUES (2, 'Frieren', 1000, 31380, 'Main');
```

```
INSERT INTO CHARACTERS (CHARACTER_ID, NAME,  
AGE, FAVORITES, ROLE)
```

```
VALUES (3, 'Brando Diego', 14, 3024, 'Main');
```

```
INSERT INTO CHARACTERS (CHARACTER_ID, NAME,  
AGE, FAVORITES, ROLE)
```

```
VALUES (4, 'Joestar Johnny', 17, 11248, 'Main');
```

```
INSERT INTO CHARACTERS (CHARACTER_ID, NAME,  
AGE, FAVORITES, ROLE)
```

```
VALUES (5, 'Elic Alphonse', 18, 15377, 'Main');
```

```
INSERT INTO CHARACTERS (CHARACTER_ID, NAME,  
AGE, FAVORITES, ROLE)
```

```
VALUES (6, 'Elic Edward', 18, 90892, 'Main');
```

```
INSERT INTO CHARACTERS (CHARACTER_ID, NAME,  
AGE, FAVORITES, ROLE)
```

```
VALUES (7, 'Denji', 18, 22888, 'Main');
```

```
INSERT INTO CHARACTERS (CHARACTER_ID, NAME,  
AGE, FAVORITES, ROLE)
```

```
VALUES (8, 'Reze', 20, 10495, 'Main');
```

Sample Table and Data

Characters Table

	CHARACTER_ID	NAME	ROLE	AGE	FAVORITES
1	1	Fern	Main	16	5663
2	2	Frieren	Main	1000	31380
3	3	Brando Diego	Main	14	3024
4	4	Joestar Johnny	Main	17	11248
5	5	Elic Alphonse	Main	18	15377
6	6	Elic Edward	Main	18	90892
7	7	Denji	Main	18	22888
8	8	Reze	Main	20	10495
9	9	Hashida Itaru	Main	19	1768
10	10	Makise Kurisu	Main	18	68581

```
CREATE TABLE characters (  
    character_id NUMBER PRIMARY KEY,  
    name VARCHAR2(200),  
    role VARCHAR2(50),  
    age NUMBER,  
    favorites NUMBER  
);
```

Physical Model 2

[Link to Physical Model Google Docs](#)

Sample Table and Data

Anime Stats Table

Query result						
Script output		DBMS output		Explain Plan		
SQL history						
Download Execution time: 0.008 seconds						
	ANIME_ID	SCORE	POPULARITY	RANKING	MEMBERS	FAVORITES
1	1	9.27	104	1	1416525	88373
2	2	9.15	1434	2	1193402	6126
3	3	9.11	3	3	3670798	241973
4	4	9.08	585	4	453669	13045
5	5	9.07	14	5	2804243	201920
6	6	9.05	20	6	2581868	63069
7	7	9.05	349	7	690353	17493
8	8	9.05	1500	8	183331	4658
9	9	9.03	8	9	3176026	228648
10	10	9.02	1742	13	152701	2566

```
CREATE TABLE anime_stats (  
    anime_id NUMBER PRIMARY KEY,  
    score NUMBER,  
    popularity NUMBER,  
    ranking NUMBER,  
    members NUMBER,  
    favorites NUMBER,  
    FOREIGN KEY (anime_id)  
    REFERENCES anime(anime_id)  
);
```

SQL Queries

```
INSERT INTO ANIME_STATS (ANIME_ID, SCORE, POPULARITY,  
RANKING, MEMBERS, FAVORITES)  
VALUES (1, 9.27, 104, 1, 1416525, 88373);
```

```
INSERT INTO ANIME_STATS (ANIME_ID, SCORE, POPULARITY,  
RANKING, MEMBERS, FAVORITES)  
VALUES (2, 9.15, 1434, 2, 1193402, 6126);
```

```
INSERT INTO ANIME_STATS (ANIME_ID, SCORE, POPULARITY,  
RANKING, MEMBERS, FAVORITES)  
VALUES (3, 9.11, 3, 3, 3670798, 241973);
```

```
INSERT INTO ANIME_STATS (ANIME_ID, SCORE, POPULARITY,  
RANKING, MEMBERS, FAVORITES)  
VALUES (4, 9.08, 585, 4, 453669, 13045);
```

```
INSERT INTO ANIME_STATS (ANIME_ID, SCORE, POPULARITY,  
RANKING, MEMBERS, FAVORITES)  
VALUES (5, 9.07, 14, 5, 2804243, 201920);
```

```
INSERT INTO ANIME_STATS (ANIME_ID, SCORE, POPULARITY,  
RANKING, MEMBERS, FAVORITES)  
VALUES (6, 9.05, 20, 6, 2581868, 63069);
```

```
INSERT INTO ANIME_STATS (ANIME_ID, SCORE, POPULARITY,  
RANKING, MEMBERS, FAVORITES)  
VALUES (7, 9.05, 349, 7, 690353, 17493);
```

```
INSERT INTO ANIME_STATS (ANIME_ID, SCORE, POPULARITY,  
RANKING, MEMBERS, FAVORITES)  
VALUES (8, 9.05, 1500, 8, 183331, 4658);
```

SQL Queries

```
INSERT INTO GENRES (GENRE_ID,  
GENRE_NAME)  
VALUES (1, 'Adventure');
```

```
INSERT INTO GENRES (GENRE_ID,  
GENRE_NAME)  
VALUES (2, 'Action');
```

```
INSERT INTO GENRES (GENRE_ID,  
GENRE_NAME)  
VALUES (3, 'Drama');
```

```
INSERT INTO GENRES (GENRE_ID,  
GENRE_NAME)  
VALUES (4, 'Comedy');
```

```
INSERT INTO GENRES (GENRE_ID,  
GENRE_NAME)  
VALUES (5, 'Award Winning');
```

```
INSERT INTO GENRES (GENRE_ID,  
GENRE_NAME)  
VALUES (6, 'Sports');
```

```
INSERT INTO GENRES (GENRE_ID,  
GENRE_NAME)  
VALUES (7, 'Fantasy');
```

```
INSERT INTO GENRES (GENRE_ID,  
GENRE_NAME)  
VALUES (8, 'Slice of Life');
```

Sample Table and Data

Genres Table

Query result	Script output	DBMS output	Explain Plan	SQL history
Download Execution time: 0.004 seconds				
	GENRE_ID	GENRE_NAME		
1	2	Action		
2	1	Adventure		
3	5	Award Winning		
4	4	Comedy		
5	3	Drama		
6	7	Fantasy		
7	8	Slice of Life		
8	6	Sports		

```
CREATE TABLE genres (  
    genre_id NUMBER PRIMARY KEY,  
    genre_name VARCHAR2(100) UNIQUE NOT  
    NULL  
);
```


Physical Model 3

[Link to Physical Model Google Docs](#)

Sample Table and Data

Anime Character Table

Query result Script output DBMS output Explain Plan SQL history				
Download Execution time: 0.009 seconds				
	ANIME_ID	CHARACTER_ID	VOICE_ACTOR_ID	
1	1	1	1	
2		1	2	2
3		2	3	3
4		2	4	4
5		3	5	5
6		3	6	6
7		4	7	7
8		4	8	8

```
CREATE TABLE anime_character (  
  anime_id NUMBER NOT NULL,  
  character_id NUMBER NOT NULL,  
  voice_actor_id NUMBER,  
  CONSTRAINT fk_ac_anime  
    FOREIGN KEY (anime_id)  
      REFERENCES anime(anime_id),  
  CONSTRAINT fk_ac_character  
    FOREIGN KEY (character_id)  
      REFERENCES characters(character_id),  
  CONSTRAINT fk_ac_voice_actor  
    FOREIGN KEY (voice_actor_id)  
      REFERENCES voice_actors(voice_actor_id),  
  CONSTRAINT pk_anime_character  
    PRIMARY KEY (anime_id, character_id)  
);
```

SQL Queries

```
INSERT INTO ANIME_CHARACTER  
(ANIME_ID, CHARACTER_ID, VOICE_ACTOR_ID)  
VALUES (1, 1, 1);
```

```
INSERT INTO ANIME_CHARACTER  
(ANIME_ID, CHARACTER_ID, VOICE_ACTOR_ID)  
VALUES (1, 2, 2);
```

```
INSERT INTO ANIME_CHARACTER  
(ANIME_ID, CHARACTER_ID, VOICE_ACTOR_ID)  
VALUES (2, 3, 3);
```

```
INSERT INTO ANIME_CHARACTER  
(ANIME_ID, CHARACTER_ID, VOICE_ACTOR_ID)  
VALUES (2, 4, 4);
```

```
INSERT INTO ANIME_CHARACTER  
(ANIME_ID, CHARACTER_ID, VOICE_ACTOR_ID)  
VALUES (3, 5, 5);
```

```
INSERT INTO ANIME_CHARACTER  
(ANIME_ID, CHARACTER_ID, VOICE_ACTOR_ID)  
VALUES (3, 6, 6);
```

```
INSERT INTO ANIME_CHARACTER  
(ANIME_ID, CHARACTER_ID, VOICE_ACTOR_ID)  
VALUES (4, 7, 7);
```

```
INSERT INTO ANIME_CHARACTER  
(ANIME_ID, CHARACTER_ID, VOICE_ACTOR_ID)  
VALUES (4, 8, 8);
```

SQL Queries

```
INSERT INTO VOICE_ACTORS (VOICE_ACTOR_ID, NAME,  
BIRTHDAY, FAVORITES, LANGUAGE)
```

```
VALUES (1, 'Ichinose Kana', '12/20/1996', 2098, 'Japanese');
```

```
INSERT INTO VOICE_ACTORS (VOICE_ACTOR_ID, NAME,  
BIRTHDAY, FAVORITES, LANGUAGE)
```

```
VALUES (2, 'Tanezaki Atsumi', '9/27/1990', 6093, 'Japanese');
```

```
INSERT INTO VOICE_ACTORS (VOICE_ACTOR_ID, NAME,  
BIRTHDAY, FAVORITES, LANGUAGE)
```

```
VALUES (3, 'Ishikawa Kaito', '10/13/1993', 9981, 'Japanese');
```

```
INSERT INTO VOICE_ACTORS (VOICE_ACTOR_ID, NAME,  
BIRTHDAY, FAVORITES, LANGUAGE)
```

```
VALUES (4, 'Sakata Shougo', '1/23/1998', 269, 'Japanese');
```

```
INSERT INTO VOICE_ACTORS (VOICE_ACTOR_ID, NAME,  
BIRTHDAY, FAVORITES, LANGUAGE)
```

```
VALUES (5, 'Kugimiya Rie', '5/30/1979', 40806, 'Japanese');
```

```
INSERT INTO VOICE_ACTORS (VOICE_ACTOR_ID, NAME,  
BIRTHDAY, FAVORITES, LANGUAGE)
```

```
VALUES (6, 'Mignogna Vic', '8/27/1962', 8163, 'English');
```

```
INSERT INTO VOICE_ACTORS (VOICE_ACTOR_ID, NAME,  
BIRTHDAY, FAVORITES, LANGUAGE)
```

```
VALUES (7, 'Toya Kikunosuke', '11/30/1998', 503,  
'Japanese');
```

Sample Table and Data

Voice Actors Table

	VOICE_ACTOR_ID	NAME	BIRTHDAY	FAVORITES	LANGUAGE
1	1	Ichinose Kana	12/20/1996	2098	Japanese
2	2	Tanezaki Atsumi	9/27/1990	6093	Japanese
3	3	Ishikawa Kaito	10/13/1993	9981	Japanese
4	4	Sakata Shougo	1/23/1998	269	Japanese
5	5	Kugimiya Rie	5/30/1979	40806	Japanese
6	6	Mignogna Vic	8/27/1962	8163	English
7	7	Toya Kikunosuke	11/30/1998	503	Japanese
8	8	Ueda Reina	1/17/1994	7597	Japanese
9	9	Seki Tomokazu	9/8/1972	6272	Japanese
10	10	Nishimura Trina	8/8/1983	682	English

```
CREATE TABLE voice_actors (  
  voice_actor_id NUMBER PRIMARY KEY,  
  name VARCHAR2(200),  
  birthday VARCHAR2(50),  
  favorites NUMBER,  
  language VARCHAR2(50)  
);
```


Physical Model 4

[Link to Physical Model Google Docs](#)

Sample Table and Data

Anime Production Table

Query result						
Download						
Execution time: 0.017 seconds						
	PRODUCTION_ID	ANIME_ID	STUDIO_ID	GENRE_ID	DEMOGRAPHIC	SOURCE
1		1	1	1	1 Shounen	Manga
2		2	2	2	2 Seinen	Manga
3		3	3	1	2 Shounen	Manga
4		4		(null)	2 Shounen	Manga
5		5	5	3	3 NULL	Visual novel
6		6	6	4	2 Shounen	Manga
7		7	7	5	2 Shounen	Manga
8		8	8	5	2 Shounen	Manga
9		9	9	6	2 Shounen	Manga
10		10	10	(null)	2 Shounen	Light novel

```
CREATE TABLE anime_production (  
    production_id NUMBER GENERATED BY  
    DEFAULT AS IDENTITY PRIMARY KEY,  
    anime_id NUMBER NOT NULL,  
    studio_id NUMBER,  
    genre_id NUMBER,  
    demographic VARCHAR2(100),  
    source VARCHAR2(100),  
    CONSTRAINT fk_anime FOREIGN KEY  
    (anime_id) REFERENCES anime(anime_id),  
    CONSTRAINT fk_studio FOREIGN KEY  
    (studio_id) REFERENCES studios(studio_id),  
    CONSTRAINT fk_genre FOREIGN KEY  
    (genre_id) REFERENCES genres(genre_id)  
);
```

SQL Queries

```
INSERT INTO ANIME_PRODUCTION  
(ANIME_ID, STUDIO_ID, GENRE_ID,  
DEMOGRAPHIC, SOURCE)  
VALUES (1, 1, 1, 'Shounen', 'Manga');  
  
INSERT INTO ANIME_PRODUCTION  
(ANIME_ID, STUDIO_ID, GENRE_ID,  
DEMOGRAPHIC, SOURCE)  
VALUES (2, 2, 2, 'Seinen', 'Manga');  
  
INSERT INTO ANIME_PRODUCTION  
(ANIME_ID, STUDIO_ID, GENRE_ID,  
DEMOGRAPHIC, SOURCE)  
VALUES (3, 1, 2, 'Shounen', 'Manga');  
  
INSERT INTO ANIME_PRODUCTION  
(ANIME_ID, STUDIO_ID, GENRE_ID,  
DEMOGRAPHIC, SOURCE)  
VALUES (4, NULL, 2, 'Shounen', 'Manga');  
  
INSERT INTO ANIME_PRODUCTION  
(ANIME_ID, STUDIO_ID, GENRE_ID,  
DEMOGRAPHIC, SOURCE)  
VALUES (5, 3, 3, 'NULL', 'Visual novel');  
  
INSERT INTO ANIME_PRODUCTION  
(ANIME_ID, STUDIO_ID, GENRE_ID,  
DEMOGRAPHIC, SOURCE)  
VALUES (6, 4, 2, 'Shounen', 'Manga');
```

SQL Queries

```
INSERT INTO STUDIOS (STUDIO_ID, STUDIO_NAME,  
STUDIO_JAPANESE_NAME, CREATION_DATE, FAVORITES)  
VALUES (1,'Aniplex','アニプレックス',1995,4446);  
  
INSERT INTO STUDIOS (STUDIO_ID, STUDIO_NAME,  
STUDIO_JAPANESE_NAME, CREATION_DATE, FAVORITES)  
VALUES (2,'Warner Bros. Japan','ワーナーブラザースジャパン',1992,125);  
  
INSERT INTO STUDIOS (STUDIO_ID, STUDIO_NAME,  
STUDIO_JAPANESE_NAME, CREATION_DATE, FAVORITES)  
VALUES (3,'Frontier Works','フロンティアワークス',2002,39);  
  
INSERT INTO STUDIOS (STUDIO_ID, STUDIO_NAME,  
STUDIO_JAPANESE_NAME, CREATION_DATE, FAVORITES)  
VALUES (4,'Production I.G','プロダクション・アイジー',1987,9817);  
  
INSERT INTO STUDIOS (STUDIO_ID, STUDIO_NAME,  
STUDIO_JAPANESE_NAME, CREATION_DATE, FAVORITES)  
VALUES (5,'TV Tokyo','テレビ東京',1964,534);  
  
INSERT INTO STUDIOS (STUDIO_ID, STUDIO_NAME,  
STUDIO_JAPANESE_NAME, CREATION_DATE, FAVORITES)  
VALUES (6,'VAP','バップ',1981,45);  
  
INSERT INTO STUDIOS (STUDIO_ID, STUDIO_NAME,  
STUDIO_JAPANESE_NAME, CREATION_DATE, FAVORITES)  
VALUES (7,'Pony Canyon','ポニーキャニオン',1966,189);
```

Sample Table and Data

Studios Table

Query result					
Download					
Execution time: 0.009 seconds					
	STUDIO_ID	STUDIO_NAME	STUDIO_JAPANESE	CREATION_DATE	FAVORITES
1		1 Aniplex	アニプレックス	1995	4446
2		2 Warner Bros. Japan	ワーナーブラザース	1992	125
3		3 Frontier Works	フロンティアワークス	2002	39
4		4 Production I.G	プロダクション・アイ	1987	9817
5		5 TV Tokyo	テレビ東京	1964	534
6		6 VAP	バップ	1981	45
7		7 Pony Canyon	ポニーキャニオン	1966	189
8		8 Shochiku	松竹	1895	35
9		9 Bandai Visual	バンダイビジュアル	1983	85
10		10 Dentsu	電通	1901	61

```
CREATE TABLE studios (  
    studio_id NUMBER PRIMARY KEY,  
    studio_name VARCHAR2(200) NOT NULL,  
    studio_japanese_name VARCHAR2(200),  
    creation_date NUMBER,  
    favorites NUMBER  
);
```

Physical Model 5

[Link to Physical Model Google Docs](#)

Sample Table and Data

Aggregate Max Function - Displaying max stats scores above 8 with name and year of anime in descending order

Query result				Script output	DBMS output	Explain Plan	SQL history
Download				Execution time: 0.009 seconds			
	MAX(SCORE)	TITLE	RELEASE_YEAR				
8	8.71	Boku no Hero Acade	2025				
9	9.08	Chainsaw Man Movie	2025				
10	8.67	Kimetsu no Yaiba Mc	2025				
11	8.67	Kingdom 6th Season	2025				
12	8.91	Kusuriya no Hitorigo	2025				
13	8.62	Luo Xiaohei Zhanji 2	2025				
14	8.68	Uma Musume Cinder	2025				
15	8.67	Bleach TYBW Souko	2024				
16	8.69	Boku no Kokoro no Y	2024				
17	8.72	Chi. Chikyuu no Und	2024				
18	8.62	Haikyuu Movie Gomi	2024				
19	8.73	Hibike! Euphonium 3	2024				
20	8.72	Kingdom 5th Season	2024				

Sample Table and Data

Join - Show Anime title and demographic where genre is "Adventure"

Query result				Script output	DBMS output	Explain Plan	SQL history
Download				Execution time: 0.005 seconds			
	TITLE	DEMOGRAPHIC	GENRE_NAME				
1	Sousou no Frieren	Shounen	Adventure				
2	Sousou no Frieren 2r	Shounen	Adventure				
3	Sen to Chihiro no Kar	NULL	Adventure				
4	Mushishi Zoku Shou	Seinen	Adventure				
5	Mushishi Zoku Shou	Seinen	Adventure				
6	Howl no Ugoku Shiro	NULL	Adventure				
7	Mushishi	Seinen	Adventure				
8	Tengen Toppa Gurrer	NULL	Adventure				
9	Luo Xiaohei Zhanji 2	NULL	Adventure				



Testing and Validation

[Link to Final Queries Google Docs](#)

Bonus Queries

```
SELECT anime.TITLE,  
       ANIME_PRODUCTION.DEMOGRAPHIC,  
       GENRES.GENRE_NAME,  
       CHARACTERS.ROLE,  
       ANIME_CHARACTER.CHARACTER_ID  
FROM ANIME  
JOIN ANIME_PRODUCTION  
  ON ANIME.ANIME_ID = ANIME_PRODUCTION.ANIME_ID  
JOIN GENRES  
  ON ANIME_PRODUCTION.GENRE_ID = GENRES.GENRE_ID  
JOIN ANIME_CHARACTER  
  ON ANIME.ANIME_ID = ANIME_CHARACTER.ANIME_ID  
JOIN CHARACTERS  
  ON ANIME_CHARACTER.CHARACTER_ID = CHARACTERS.CHARACTER_ID  
WHERE GENRE_NAME = 'Adventure';
```

Query result

Script output

DBMS output

Explain Plan

SQL history

Download

Execution time: 0.009 seconds

	TITLE	DEMOGRAPHIC	GENRE_NAME	ROLE	CHARACTER_ID
1	Made in Abyss	NULL	Adventure	Main	144
2	Made in Abyss	NULL	Adventure	Main	145
3	Sousou no Frieren	Shounen	Adventure	Main	1
4	Sousou no Frieren	Shounen	Adventure	Main	2
5	Sousou no Frieren 2r	Shounen	Adventure	Main	1
6	Sousou no Frieren 2r	Shounen	Adventure	Main	2
7	Sen to Chihiro no Kar	NULL	Adventure	Main	65
8	Sen to Chihiro no Kar	NULL	Adventure	Main	66
9	Mushishi Zoku Shou	Seinen	Adventure	Main	86
10	Mushishi Zoku Shou	Seinen	Adventure	Supporting	87
11	Mushishi Zoku Shou	Seinen	Adventure	Main	86

```
SELECT ANIME.TITLE, CHARACTERS.NAME, CHARACTERS.ROLE,  
       ANIME_CHARACTER.CHARACTER_ID, ANIME_PRODUCTION.SOURCE,  
       GENRES.GENRE_NAME  
FROM ANIME  
JOIN ANIME_CHARACTER  
  ON ANIME.ANIME_ID = ANIME_CHARACTER.ANIME_ID  
JOIN CHARACTERS  
  ON CHARACTERS.CHARACTER_ID = ANIME_CHARACTER.CHARACTER_ID  
JOIN ANIME_PRODUCTION  
  ON ANIME_PRODUCTION.ANIME_ID = ANIME_CHARACTER.ANIME_ID  
JOIN GENRES  
  ON GENRES.GENRE_ID = ANIME_PRODUCTION.GENRE_ID  
WHERE CHARACTERS.ROLE = 'Supporting'  
ORDER BY ANIME_CHARACTER.CHARACTER_ID;
```

Query result		Script output	DBMS output	Explain Plan	SQL history	
		Download ▾	Execution time: 0.015 seconds			
	TITLE	NAME	ROLE	CHARACTER_ID	SOURCE	GENRE_NAME
1	Mushishi Zoku Shou	Akane	Supporting	87	Manga	Adventure
2	Mushishi Zoku Shou	Adashino	Supporting	105	Manga	Adventure
3	Mushishi	Adashino	Supporting	105	Manga	Adventure
4	Violet Evergarden	Abelfreyja Drossel Cr	Supporting	106	Light novel	Drama
5	Rurouni Kenshin Tsui	Hijikata Toshizo	Supporting	108	Manga	Action
6	Odd Taxi	Baba Atsuya	Supporting	139	Original	Award Winning



Conclusion and Q&A



Summary

Overall, this project presents a structured relational database for managing anime information, including titles, characters, voice actors, studios, genres, and production details. The design ensures that the data is organized efficiently, supports complex relationships, and enables easy retrieval of information such as character roles, voice actors, and anime statistics. And by applying normalization up to 3NF to the database, the redundancy is reduced, ensuring consistency across all entities.